Worked example: from input to recommendation

Generated by `python -m notebooks.phase5_walkthrough` from the data and `config.yaml`; re-running it gives the same numbers. Example: **A: $10,000, medium risk**, as of **2026-08-31**.

1. Input

The user gives two things: **$10,000** and risk tolerance **medium**. The risk tolerance sets the risk-aversion parameter **γ = 5** (low / medium / high = 10 / 5 / 2). Everything else is fixed by the project, not chosen by the user (advisor feedback A4): the 51-stock universe, long-only, at most **10% per stock**, and the sector rule, **no sector above 25%**, the same constraints the backtest uses.

2. Data as of 2026-08-31

Only data dated on or before 2026-08-31 is used (tested by scrambling everything after it). Σ is the sample covariance of the last 252 trading days of daily returns, scaled to a month. The risk-free rate is 3.73% a year.

3. ML forecasts (the views Q)

The Phase 4 headline model, the elastic net, is retrained with the Phase 3 walk-forward on data up to 2026-08-31 and forecasts each stock's next-month excess return. The forecasts range from 0.80% to 1.51% a month (51 distinct values across 51 stocks).

4. Black–Litterman: prior, view uncertainty, posterior

- **Prior Π = δΣw_mkt** (δ = 2.5): the returns that make today's market-cap weights optimal.
- **View uncertainty Ω-A**: each stock's mean squared past forecast error (last 24 months), shrunk toward its return variance. Large Ω = little trust.
- **Posterior μ_BL** (T6), τ = 0.05: μ_BL moved **31%** of the way from the prior toward the views (‖μ_BL − Π‖ / ‖Q − Π‖), so the weak forecasts change little.

Monthly %, the 5 highest and 5 lowest μ_BL:

ticker	sector	w_mkt	Pi	Q	omega_sd	mu_bl
AMZN	Consumer Discretionary	8.71	0.65	1.45	9.51	0.80
GOOGL	Communication Services	12.90	0.61	1.43	10.20	0.76
SHW	Materials	0.26	0.26	1.34	7.22	0.74
NEM	Materials	0.41	0.45	1.47	13.09	0.72
LOW	Consumer Discretionary	0.36	0.20	1.18	7.04	0.70
SO	Utilities	0.31	-0.10	0.84	4.34	0.17
DUK	Utilities	0.29	-0.11	0.80	3.99	0.14
COP	Energy	0.49	-0.16	0.91	8.38	-0.05
CVX	Energy	1.26	-0.14	0.88	7.82	-0.05
XOM	Energy	2.06	-0.17	0.88	7.09	-0.07

(w_mkt = market-cap weight; omega_sd = √Ω, the view's standard error.)

5. Optimization

maximize $\mu_{BL}^{T}w - (\gamma/2)w^{T}\Sigma w$ subject to $1^{T}w = 1$, $0 \le w_i \le 10\%$, every sector $\le 25\%$.

- Solver status **optimal**; **49 stocks** in the target portfolio.
- At the 10% stock cap: none.
- Sectors at the 25% cap: none (largest sector: Information Technology, 23.3%).
- KKT check (T4), largest violation: stationarity 4.4e-18, primal 2.2e-16, dual 0.0e+00, complementarity 3.0e-10, so the solution is the certified optimum.

6. Whole shares

Target weight × $10,000 ÷ the closing price on 2026-08-31, rounded down; the leftover cash then buys single shares only where that brings a holding closer to its target, never breaking a cap. Result: **$9,473.79 invested in 36 stocks, $526.21 cash**; the largest gap between a held and a target weight is 1.47%.

ticker	sector	price	shares	value	weight	target
GOOGL	Communication Services	339.35	2	678.70	6.8	6.6
NVDA	Information Technology	220.78	3	662.34	6.6	7.6
AAPL	Information Technology	316.85	2	633.70	6.3	7.8
AMZN	Consumer Discretionary	259.77	2	519.54	5.2	4.6
MSFT	Information Technology	507.29	1	507.29	5.1	5.7
JPM	Financials	356.02	1	356.02	3.6	3.8
D	Utilities	66.01	5	330.05	3.3	3.6
NEE	Utilities	82.34	4	329.36	3.3	2.9
HD	Consumer Discretionary	327.83	1	327.83	3.3	2.0
WMT	Consumer Staples	104.87	3	314.61	3.1	2.9
PSA	Real Estate	306.99	1	306.99	3.1	2.5
UNP	Industrials	300.67	1	300.67	3.0	2.2
MRK	Health Care	147.76	2	295.52	3.0	2.3
PLD	Real Estate	139.92	2	279.84	2.8	2.2
JNJ	Health Care	265.85	1	265.85	2.7	1.7
SO	Utilities	88.00	3	264.00	2.6	2.3
MCD	Consumer Discretionary	263.54	1	263.54	2.6	1.9
WFC	Financials	86.39	3	259.17	2.6	2.7
ABT	Health Care	110.36	2	220.72	2.2	1.8
HON	Industrials	213.53	1	213.53	2.1	1.6
LOW	Consumer Discretionary	204.70	1	204.70	2.0	1.8
SLB	Energy	60.10	3	180.30	1.8	1.9
AMT	Real Estate	175.70	1	175.70	1.8	1.4
PFE	Health Care	28.46	6	170.76	1.7	1.7
XOM	Energy	160.95	1	160.95	1.6	1.7
VZ	Communication Services	50.02	3	150.06	1.5	1.4
PG	Consumer Staples	145.12	1	145.12	1.5	1.1

ticker	sector	price	shares	value	weight	target
PEP	Consumer Staples	140.34	1	140.34	1.4	2.1
CMCSA	Communication Services	26.62	5	133.10	1.3	1.5
COP	Energy	132.49	1	132.49	1.3	1.3
CSCO	Information Technology	110.49	1	110.49	1.1	1.3
DIS	Communication Services	107.55	1	107.55	1.1	1.3
UPS	Industrials	104.23	1	104.23	1.0	0.6
KO	Consumer Staples	88.67	1	88.67	0.9	1.0
NKE	Consumer Discretionary	39.06	2	78.12	0.8	0.8
BAC	Financials	61.94	1	61.94	0.6	0.7

(weight and target in %; weight = value ÷ $10,000, so cash is unallocated.)

7. Checks

- Constraints, target weights: long_only ✓, cap ✓, sector_max ✓, budget ✓.
- Constraints, whole shares: long_only ✓, cap ✓, sector_max ✓.
- **Same as the backtest:** the tool's target weights equal the weights the backtested S6 strategy holds from 2026-08-31 (largest difference 0.0e+00).
- Reproducible: a second run gives identical holdings and summary (✓).
- Expected volatility 9.7% a year (target portfolio 10.2%); model expected excess return 5.9% a year, a model estimate, not a promise.

Sector weights (%), target and held, against the 25% rule:

sector	target	held	max
Information Technology	23.3	19.1	25
Consumer Discretionary	11.1	13.9	25
Communication Services	10.7	10.7	25
Financials	9.4	6.8	25
Utilities	8.8	9.2	25
Consumer Staples	8.3	6.9	25
Health Care	8.2	9.5	25
Real Estate	6.2	7.6	25
Industrials	5.2	6.2	25
Energy	4.9	4.7	25
Materials	4.0	0.0	25